

Notes: Shue & Townsend (JF, 2017)

How Do Quasi-Random Option Grants Affect CEO Risk-Taking?

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1 Big picture

- **Question.** Does giving a CEO more new at-the-money stock options cause the CEO to take more risk?
- **Why the question is hard.** The theoretical sign is ambiguous for a risk-averse CEO. More options raise **vega**, which strengthens the convexity incentive to take risk, but they also raise **delta**, which makes the CEO's wealth more sensitive to the stock price and can push in the opposite direction.
- **Why existing evidence is not enough.** Firms do not randomly choose option grants. A board may raise option pay exactly when the firm wants more risk-taking, so ordinary correlations between options and risk are hard to interpret causally.
- **Core empirical idea.** The paper exploits **multiyear option-grant plans**. These plans generate quasi-random timing in when large option increases occur, and they also generate quasi-random differences in how option value responds to stock-price movements.
- **Data.** The main sample uses **Execucomp CEO grant-level compensation data from 1992 to 2010**, combined with **Compustat**, **CRSP**, and Fama-French industry returns.
- **IV strategy 1.** For CEOs on **fixed-value plans**, option value tends to jump at the start of a new cycle. The instrument is an indicator for whether a CEO-year is **predicted** to be the first year of a new fixed-value cycle.
- **IV strategy 2.** For CEOs on **fixed-number plans**, the Black-Scholes value of a new at-the-money grant rises with the stock price. The instrument is the interaction between a **fixed-number-plan indicator** and **industry returns** within a cycle.
- **Main first-stage fact.** Predicted first years raise the log Black-Scholes value of new grants by about **0.14 to 0.15 log points**, and the fixed-number $\times$ industry-return interaction is also very strong, with first-stage F -statistics well above conventional weak-instrument thresholds.
- **Main result.** A **10% increase in new option grants** leads to about a **2.8% to 4.2% increase in equity volatility**, depending on the instrument and volatility window used.
- **Main mechanism.** The increase in risk is driven largely by **higher leverage**. Roughly **45% to 46%** of the volatility increase can be mechanically explained by the induced increase in leverage.
- **What does not move much.** Investment responds weakly. Capital expenditures and total investment increase only modestly and are usually statistically insignificant in the baseline specifications.
- **Key heterogeneity.** The effect is stronger when the new option grant is **large relative to the CEO's existing unexercised option portfolio**. There is also suggestive evidence of larger effects in **finance and high-tech** firms.
- **Economic takeaway.** In the range of option grants commonly observed in large public firms, giving CEOs more new at-the-money options makes them take more risk on average, mainly by leveraging the firm up.

2 Data and measurement (key objects)

2.1 Data sources and baseline sample

- Executive compensation data come from **Execucomp** over **1992–2010**.
- The analysis focuses on **CEOs**, not all top executives.
- Grant-level option data are used so that the authors can recover the structure of multiyear plans and identify the exact timing of grants.
- Accounting variables come from **Compustat**.
- Stock returns and volatility come from **CRSP**.
- Industry returns come from the **Fama-French 49-industry classification**.
- For tests using accounting outcomes, **financial firms** (SIC 6000 to 6999) and **regulated utilities** (SIC 4800 to 4999) are excluded.

Interpretation. This is a standard large-firm public-company sample, which makes the paper directly relevant for CEO pay design at S&P 1500-type firms.

2.2 Why grant-level option data matter

- The authors need the **date**, **number**, and **value** of each option grant to detect compensation cycles.
- Exact grant dates are directly available after **2006**.
- Before 2006, firms generally disclosed only the **expiration date**; the paper infers the grant date by assuming that options expire on the grant-date anniversary.
- Starting in 2006, firms report grant-date option values directly; before 2006, the paper uses Execucomp's computed **Black-Scholes value**.

Interpretation. The whole design depends on measuring the timing and size of new grants accurately, not just annual totals of compensation.

2.3 How fixed-number cycles are detected

- A CEO is treated as being on a **fixed-number cycle** if the CEO receives the **same number of options** in consecutive years, adjusting for stock splits.
- When a CEO receives multiple grants in one year, the paper compares the **largest grant** across years.
- To ensure that the detected cycle is economically meaningful, the candidate fixed-number grants must account for more than **50%** of the total options granted over the cycle years.

Interpretation. A fixed-number plan means the board mechanically keeps the *number* of options unchanged from year to year inside the cycle.

2.4 How fixed-value cycles are detected

- A CEO is treated as being on a **fixed-value cycle** if the value of options granted is within **3%** of the previous year's value.
- The value benchmark can be either **Black-Scholes value** or **face value** (number of options times grant-date stock price).
- The same valuation method must be used consistently inside the cycle.
- The 3% tolerance is allowed because firms often grant options in round lots, so value is not literally identical every year.

Interpretation. A fixed-value plan means the board mechanically keeps the *dollar value* of option compensation roughly unchanged inside the cycle.

2.5 Cycle-length facts

- The paper identifies **4,506 CEO-year observations** on fixed-number cycles and **2,977 CEO-year observations** on fixed-value cycles.
- For fixed-number plans, the modal cycle length is **two years**: **61.14%** of observations are on two-year cycles, **22.95%** on three-year cycles, and the rest on longer cycles.
- For fixed-value plans, the concentration is even stronger: **89.55%** of observations are on two-year cycles and **9.37%** on three-year cycles.
- In a representative cross-section for fiscal year 2000, the sample contains **270 fixed-number CEO-years**, **144 fixed-value CEO-years**, and **1,165 other CEO-years** that do not fit either pattern.

Interpretation. Two-year cycles dominate. That regularity is exactly what allows the authors to forecast when new cycles should start.

2.6 Option-compensation variables

- The main endogenous regressor is the annual change in the log **Black-Scholes value of new option grants**.
- The paper also studies annual changes in the **delta** and **vega** of the new grants.
- For prior holdings, the authors compute the change in the log **total outstanding delta** and **total outstanding vega** of previously granted stock and unexercised options.

Interpretation. The paper is not asking whether higher *vega alone* matters. It asks what happens when a CEO gets *more at-the-money options*, which jointly raises grant value, delta, and vega.

2.7 Risk and mechanism outcomes

- The main outcome is realized **equity volatility**.
- The first volatility measure is the annualized volatility of daily returns over the **12 months after the grant date**.
- The second is the annualized volatility of daily returns over the first **120 trading days after the grant date**.
- Mechanism outcomes include **market leverage**, **log total debt**, and an indicator for whether **net debt issuance is positive**.
- Real-side outcomes include **log capital expenditures** and **log total investment**, where total investment sums capital expenditures, R&D, acquisitions, and advertising.

Interpretation. Volatility is the natural reduced-form outcome because option value depends directly on volatility. Leverage and investment are then used to understand where the volatility response comes from.

2.8 Controls and estimation sample sizes

- All main regressions include **year fixed effects**.
- Standard errors are **clustered by firm**.
- Controls include lagged CEO tenure, log cash compensation, log sales, log assets, sales growth, market-to-book, tangibility, and an indicator for rated debt.
- The baseline **IV1** volatility sample contains about **3,692 CEO-year observations**.
- The baseline **IV2** volatility sample contains about **3,535 CEO-year observations**.
- Accounting-outcome specifications are smaller, roughly **2,815** observations for IV1 and **2,930** for IV2.

Interpretation. The paper has enough staggered firm-year variation to run credible IV designs, but the identification comes from a fairly specific subset of CEOs on multiyear plans.

3 Models and empirical specifications (all equations + interpretation)

3.1 Theoretical ambiguity: convexity effect versus magnification effect

For a new at-the-money option grant, increasing the number of options raises three objects at once:

$$\text{Grant Value}, \quad \Delta(\text{delta}), \quad \text{and} \quad \nu(\text{vega}). \quad (1)$$

The paper’s key conceptual point is that these objects push risk-taking in different ways.

- Higher **vega** strengthens the payoff convexity of compensation and tends to encourage more risk-taking.
- Higher **delta** raises the sensitivity of CEO wealth to the stock price and can make a risk-averse CEO want to reduce risk.

Interpretation. The total effect of “more options” is not mechanically positive. The sign is an empirical question.

3.2 IV strategy 1: predicted first years of fixed-value cycles

For CEOs on fixed-value plans, the first-stage regression is

$$\Delta O_{ijt} = \beta_0 + \beta_1 \text{PredictedFirstYear}_{ijt} + \gamma_t + X'_{ijt-1}\beta + \varepsilon_{ijt}, \quad (2)$$

and the second stage is

$$\Delta Y_{ijt} = \delta_0 + \delta_1 \widehat{\Delta O}_{ijt} + \gamma_t + X'_{ijt-1}\delta + u_{ijt}. \quad (3)$$

Here:

- ΔO_{ijt} is the annual change in the log Black-Scholes value of the new option grant,
- ΔY_{ijt} is the annual change in an outcome such as volatility or leverage,
- γ_t are year fixed effects, and
- X_{ijt-1} is the lagged control set.

Interpretation. The instrument does not use the actual size of the pay jump. It uses the staggered timing of when option compensation is *predicted* to jump on average.

3.3 Prediction rule for cycle first years

The paper predicts the first year of a new fixed-value cycle using only past information. Let:

- k be the length of the CEO’s last completed fixed-value cycle, and
- n_t be the number of consecutive years up through year t in which option value has remained fixed within the allowed tolerance.

Then year $t + 1$ is predicted to be the first year of a new cycle if

$$n_t \geq k. \quad (4)$$

If the CEO has no prior completed cycle, the paper sets $k = 2$, the modal cycle length in the data.

Interpretation. This replaces actual cycle starts, which may be renegotiated endogenously, with mechanically forecasted cycle starts.

3.4 IV strategy 2: fixed-number plans interacted with industry returns

Within a fixed-number cycle, a CEO receives the same number of at-the-money options each year. Because the Black-Scholes value of one at-the-money option scales with the stock price, option value moves with returns inside the cycle. The first-stage equation is

$$\Delta O_{ijt} = \beta_0 + \beta_1 FN_{ijt} + \beta_2 R_{kt} + \beta_3 (FN_{ijt} \times R_{kt}) + \gamma_t + X'_{ijt-1} \beta + \varepsilon_{ijt}, \quad (5)$$

and the second stage is

$$\Delta Y_{ijt} = \delta_0 + \delta_1 FN_{ijt} + \delta_2 R_{kt} + \delta_3 \widehat{\Delta O}_{ijt} + \gamma_t + X'_{ijt-1} \delta + u_{ijt}. \quad (6)$$

Here FN_{ijt} is an indicator for a fixed-number plan and R_{kt} is the CEO firm's **industry return over the prior 12 months**.

Interpretation. The excluded instrument is not industry returns by themselves, but the **differential** sensitivity of fixed-number CEOs' option grants to those returns.

3.5 Why the analysis is done in changes rather than levels

- The instruments shift the **timing of increases** in new option grants.
- Therefore the natural outcome is the **change** in risk-taking around the grant, not the level of risk.
- Stock variables such as volatility and leverage are modeled as annual changes.
- Flow variables such as capital expenditures and total investment are used in levels over the following year.

Interpretation. The design asks whether risk increases more in years when option compensation jumps exogenously than in otherwise similar years.

3.6 What parameter is identified

- The estimated coefficient captures the effect of a change in **new at-the-money option grants**.
- It is therefore the **overall net effect** of simultaneously changing grant value, delta, and vega.
- It is *not* the effect of changing vega while holding delta fixed, and it is not the effect of the CEO's entire lifetime option portfolio.

Interpretation. This is exactly the policy-relevant parameter for a compensation committee deciding whether to grant a CEO more or fewer at-the-money options this year.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why IV1 is credible

- Fixed-value CEOs cannot receive upward drift in option value *within* a cycle, so when compensation trends upward over time, increases tend to arrive in discrete jumps at the beginning of new cycles.
- These cycle starts are **staggered across firms**, which makes them usable in a year-fixed-effects design.
- The paper uses **predicted** first years rather than actual first years, which helps break the link between the instrument and endogenous renegotiation of cycle timing.
- The identifying assumption is that firms do not set the *length of the current cycle* in anticipation of risk-taking conditions when the next cycle is forecast to begin.

Interpretation. The instrument exploits quasi-random timing, not arbitrary cross-sectional differences in pay levels.

4.2 Why IV2 is credible

- Within fixed-number cycles, industry-return shocks mechanically move the value of new grants.
- Within fixed-value cycles, the board offsets those price movements by changing the number of options so that grant value stays roughly fixed.
- The identifying restriction is that fixed-number and fixed-value firms do **not** differ in their *noncompensation-induced* response to industry returns.
- The paper does *not* require fixed-number and fixed-value firms to have the same average level of risk or the same exposure to industry returns in levels; those main effects are explicitly controlled for.

Interpretation. The identifying variation is a difference-in-slope design: does risk move more with returns for fixed-number firms only because their option grants move more with returns?

4.3 First-stage strength

- In **Table II**, the predicted-first-year instrument raises the log Black-Scholes value of new options by about **0.144 log points** in the specification with controls.
- The same instrument also raises the log delta and log vega of new grants by roughly **0.165** and **0.161** log points, respectively.
- In **Table VII**, the interaction of fixed-number status with industry returns strongly predicts changes in option value, with coefficients around **0.55** on the interaction term.

- First-stage F -statistics are generally between roughly **40** and **140**, well above standard weak-instrument concerns.

Interpretation. The instruments are not just conceptually appealing; they also move compensation enough to produce economically meaningful first stages.

4.4 Direct tests of the exclusion restriction

- The predicted-first-year instrument has no meaningful relation to changes in **product-announcement activity**, **business-expansion announcements**, **debt-maturity cycles**, or **CEO turnover**.
- When the predicted-first-year instrument is partitioned into cases where the Black-Scholes value of options actually **rises** versus **does not rise**, only the rise group shows a clear volatility response.
- For IV2, the paper allows for **nonlinear** effects of industry returns by using a five-degree polynomial and interactions with controls; the results remain similar.
- Placebo tests using firms that stop granting options around **FAS123r** show no analogous response when the option channel is shut down.

Interpretation. These tests make it harder to argue that the instruments are picking up some other latent cycle of firm activity unrelated to option grants.

4.5 Compensation and backdating robustness

- The results survive controls for **total compensation**.
- They also survive separate controls for changes in **salary**, **bonus**, **restricted stock**, and other compensation.
- Restricting the sample to years with **no restricted stock grants** yields similar results.
- The results are similar when the sample is restricted to **two-year cycles only** or to CEO-years with **one option grant only**.
- Using **predicted grant dates** rather than reported grant dates leaves the volatility results intact, which weakens the concern that strategic grant timing or backdating drives the findings.

Interpretation. The estimated effect appears tied specifically to option grants, not to a broader shock to the whole compensation package.

4.6 What the design can and cannot say

- The design identifies the effect of **moderate increases in new option grants** in the range commonly observed in large public firms.

- It does not identify the effect of extremely large option grants that might push very risk-averse, undiversified CEOs into a different region of the utility function.
- The cleanest causal evidence comes from firms using **multiyear plans**. The paper provides suggestive matching evidence that the results generalize beyond that subset, but that part is weaker than the core IV evidence.

Interpretation. The paper’s claim is intentionally local and policy-relevant: for the kinds of option increases boards usually consider, risk-taking rises.

5 Tables and figure

5.1 Table I and Figure 1: the cycle structure is real and economically meaningful

Table I
Summary Statistics

Panel A of this table shows the distribution of cycle length, with observations at the CEO-year level. Panel B shows the industry distribution, broken down by the CEO's plan type. Industries are categorized using the Fama-French 12-industry classification scheme. Panel C compares other firm and CEO characteristics across cycle types, showing the 25th, 50th, and 75th percentiles of the distributions. Because there are time trends in the prevalence of fixed-number and fixed-value cycles, we do not pool all years. Panels B and C show only summary statistics from fiscal year 2000. Fiscal years 1995 and 2005 are shown in the Internet Appendix.

Panel A: Length of Cycles					
	Fixed Number		Fixed Value		
	Freq	Percent	Freq	Percent	
2	2,755	61.14	2,666	89.55	
3	1,054	22.95	279	9.37	
4	387	8.59	18	0.60	
5+	330	7.32	14	0.47	
Total	4,506	100.00	2,977	100.00	

Panel B: Industry Distribution						
Year: 2000	Fixed Number		Fixed Value		Other	
	Freq	Percent	Freq	Percent	Freq	Percent
Consumer Nondurables	17	6.30	8	5.56	74	6.35
Consumer Durables	9	3.33	7	4.86	29	2.49
Manufacturing	36	13.33	29	20.14	116	9.96
Energy	10	3.70	6	4.17	52	4.46
Chemicals	10	3.70	5	3.47	31	2.65
Business Equipment	45	16.67	15	10.42	265	22.75
Telecommunications	8	2.96	3	2.08	36	3.09
Utilities	15	5.56	10	6.94	56	4.81
Shops	28	10.37	21	14.58	146	12.53
Health	25	9.36	7	4.86	85	7.30
Finance	33	12.22	21	14.58	148	12.70
Other	34	12.59	12	8.33	127	10.90
Total	270	100.00	144	100.00	1,165	100.00

(Continued)

Table I—Continued
Panel C: Other Characteristics

Year: 2000	Fixed Number			Fixed Value			Other		
	p25	p50	p75	p25	p50	p75	p25	p50	p75
Firm Level:									
Assets (Millions)	563.47	1,545.80	6,393.29	750.29	2,051.62	6,300.42	444.27	1,383.17	5,036.34
Sales (Millions)	476.09	1,202.65	3,211.77	640.98	1,496.26	3,994.24	372.11	1,095.55	3,708.53
Market to Book	1.09	1.42	2.35	1.09	1.37	2.38	1.12	1.54	2.53
Volatility (12 Months)	0.35	0.46	0.71	0.31	0.44	0.58	0.37	0.51	0.76
Volatility (120 Trading Days)	0.38	0.50	0.72	0.39	0.49	0.59	0.43	0.55	0.79
CAPX/PPE	0.14	0.23	0.42	0.13	0.23	0.38	0.15	0.26	0.51
Acquisitions (Millions)	0.00	1.01	48.53	0.00	0.00	39.97	0.00	0.66	38.30
Market Leverage	0.06	0.22	0.44	0.06	0.21	0.43	0.03	0.17	0.39
Book Leverage	0.19	0.41	0.57	0.19	0.42	0.57	0.10	0.37	0.55
Total Dividends (Millions)	0.00	3.05	37.73	0.00	11.45	62.96	0.00	0.00	30.41
Return on Assets	0.10	0.15	0.22	0.10	0.15	0.23	0.08	0.15	0.24
Cash Flow/Assets	0.07	0.10	0.17	0.08	0.10	0.16	0.06	0.11	0.18
CEO Level:									
Salary (Thousands)	385.00	567.12	834.00	400.00	627.60	833.65	321.16	500.00	754.65
Bonus (Thousands)	184.00	440.00	940.83	182.91	416.25	835.04	79.47	315.00	800.00
Number New Options	50.00	105.00	225.00	50.00	117.17	350.00	80.00	180.00	438.75
Number Prev Options	228.29	475.00	1,022.50	207.56	433.43	975.96	128.75	375.00	855.00
Value New Options (Thousands)	572.43	1,278.20	3,556.43	652.36	1,187.33	3,036.36	892.31	2,417.57	6,631.91
Value Prev Options (Thousands)	1,870.69	5,624.94	17,512.38	1,763.59	4,556.15	13,947.44	788.34	4,129.09	13,884.27
Delta New Options	7.91	18.16	52.67	8.82	17.66	45.51	10.91	31.04	83.41
Delta Prev Options	30.62	84.67	242.06	32.72	75.15	211.23	12.29	61.22	192.95
Vega New Options	5.86	13.80	38.56	7.39	14.14	38.60	6.87	19.97	51.48
Vega Prev Options	17.54	43.61	125.54	23.52	45.15	122.39	6.17	24.62	74.79
Value Prev Options + Stock	2,150.92	6,519.80	20,823.43	2,185.94	5,197.37	14,455.02	949.50	4,752.08	14,484.78
Delta Prev Options + Stock	31.66	89.07	258.88	34.33	76.80	228.73	14.59	67.40	197.01
Observations	270			144			1,165		

Read:

- Two-year cycles dominate for both fixed-number and fixed-value plans, especially the latter.
- In fiscal year 2000, fixed-number and fixed-value firms look broadly similar in size and leverage, while the “other” firms are more heterogeneous.
- Figure 1 shows three real fixed-value examples where option value follows an increasing step function and cycle starts are naturally staggered across CEOs.

Economic message. The identifying variation is not manufactured by the econometrics. It comes from a visible institutional pattern in how boards actually grant options.

5.2 Table II to Table V: IV1 delivers the main result inside the fixed-value sample

Read:

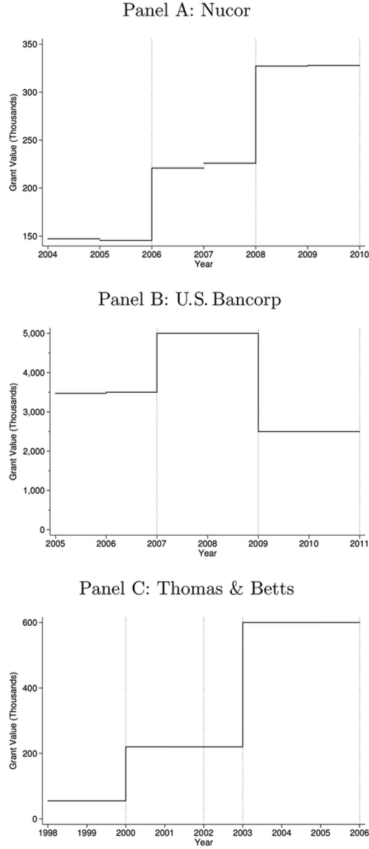


Figure 1. Real examples of fixed-value cycles and predictions. This figure shows three examples of fixed-value cycles taken from the data. Years that we predict to be cycle first years are indicated by dotted vertical lines.

Table II
IV1: First Stage

Panel A of this table shows how option compensation changes in fixed-value cycle first years. Panel B shows how option compensation changes in fixed-value cycle *predicted* first years. Observations are at the CEO-year level. The sample is limited to CEOs who are currently on fixed-value cycles or were in the previous year. *First Year* is an indicator variable equal to one in the year following the final year of a cycle. *Predicted First Year* is an indicator variable equal to one if the year is predicted to be a cycle first year based on the length of the previous cycle (see Section II.A for a detailed discussion of our prediction methodology). The variable *B-S Value* equals the Black-Scholes value of new option compensation, *Delta* equals the change in the Black-Scholes value of new option compensation associated with a 1% change in the price of the underlying, and *Vega* equals the change in the Black-Scholes value of new option compensation associated with a 0.01 change in the annualized volatility of the underlying. Control variables include CEO tenure, log of cash compensation (salary + bonus), log sales, log assets, sales growth, market to book, tangibility ratio, and a dummy variable for whether the firm has rated debt. These control variables are measured in the year prior to the year of the current option grant. In addition, controls for the change in the log delta and vega of previously granted (unexercised) option and stock holdings are included. That is, we calculate the log of total outstanding delta and vega as of the current grant date (excluding the current grant) and subtract from this the log of total outstanding delta and vega following the previous year's grant. Standard errors appear in parentheses and are clustered by firm. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Real First Years						
	$\Delta \text{ Log B-S Value}$		$\Delta \text{ Log Delta}$		$\Delta \text{ Log Vega}$	
	(1)	(2)	(3)	(4)	(5)	(6)
First Year	0.149*** (0.0193)	0.152*** (0.0192)	0.165*** (0.0265)	0.176*** (0.0256)	0.157*** (0.0312)	0.171*** (0.0302)
Controls	No	Yes	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.084	0.097	0.032	0.088	0.105	0.153
Observations	3,692	3,692	3,692	3,692	3,692	3,692

Panel B: Predicted First Years						
	$\Delta \text{ Log B-S Value}$		$\Delta \text{ Log Delta}$		$\Delta \text{ Log Vega}$	
	(1)	(2)	(3)	(4)	(5)	(6)
Predicted First Year	0.141*** (0.0185)	0.144*** (0.0185)	0.152*** (0.0255)	0.165*** (0.0244)	0.146*** (0.0296)	0.161*** (0.0286)
Controls	No	Yes	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.082	0.095	0.031	0.087	0.104	0.152
Observations	3,692	3,692	3,692	3,692	3,692	3,692

Table III
Sensitivity of New Grants to Stock Price: Fixed Value versus Fixed Number

This is a simple example adapted from Hall (1999) to illustrate how the Black-Scholes value of new at-the-money option grants and the number of options granted varies with stock price fluctuations for executives on fixed-number and fixed-value plans. For illustrative purposes, we assume the annual standard deviation is 32%, the risk-free rate is 6%, the dividend rate is 3%, and the maturity is 10 years.

Plan		Stock Price		
		Year 1 Grant 100	Year 2 Grant 120	Year 3 Grant 144
Fixed Value	Value of Options	\$1,000,000	\$1,000,000	\$1,000,000
	Number of Options	28,128	23,440	18,752
Fixed Number	Value of Options	\$1,000,000	\$1,000,000	\$1,000,000
	Number of Options	28,128	28,128	28,128

- **Table II:** predicted first years raise the log Black-Scholes value of new grants by about **0.144**, with similar increases in delta and vega.
- **Table IV:** the IV coefficient of option grants on the change in 12-month volatility is about **0.110**; for 120-trading-day volatility it is about **0.098**. A 10% increase in new option grants implies roughly a **3.8%** increase in volatility relative to the sample median.
- **Table V:** the same option shock raises market leverage, log debt, and the probability of positive net debt issuance. The book-debt effect is about **0.129**, meaning a 10% increase in options raises debt by about **1.3%**. Investment effects are small and imprecisely estimated.

Table IV
IV1: Volatility

Panel A of this table shows instrumental variable estimation results, where the variable $\Delta \text{Log } B\text{-}S \text{ Value}$ is instrumented using the *Predicted First Year* indicator, as defined in Table II. Observations are at the CEO-year level. The sample is limited to CEOs who are currently on fixed-value cycles or were in the previous year. We measure volatility in two ways: (1) the annualized volatility of daily returns in the first 12 months following the grant date, and (2) the annualized volatility of daily returns in the first 120 trading days following the grant date. Panel B shows the results of the reduced-form estimation in which these outcomes are regressed directly on the *Predicted First Year* instrument. Control variables are those listed in Table II. Standard errors appear in parentheses and are clustered by firm. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Instrumental Variable Estimation				
	Δ 12-Month Volatility		Δ 120-TD Volatility	
	(1)	(2)	(3)	(4)
$\Delta \text{ Log } B\text{-}S \text{ Value}$	0.115*** (0.0390)	0.110*** (0.0380)	0.108*** (0.0378)	0.0978*** (0.0368)
Controls	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes
F-stat. (First Stage)	58.29	60.76	58.29	60.76
Observations	3,692	3,692	3,692	3,692
Panel B: Reduced-Form Estimation				
	Δ 12-Month Volatility		Δ 120-TD Volatility	
	(1)	(2)	(3)	(4)
Predicted First Year	0.0163*** (0.00515)	0.0158*** (0.00518)	0.0152*** (0.00500)	0.0141*** (0.00504)
Controls	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes
R ²	0.300	0.302	0.279	0.284
Observations	3,692	3,692	3,692	3,692

Table V
IV1: Other Outcomes

Panel A shows instrumental variable estimation results, where the variable $\Delta \text{Log } B\text{-}S \text{ Value}$ is instrumented using the *Predicted First Year* indicator, as defined in Table II. Observations are at the CEO-year level. The sample is limited to CEOs who are currently on fixed-value cycles or were in the previous year. Financial firms (SIC 6000 to 6999) and regulated utilities (SIC 4800 to 4999) are excluded. The variable *Lev Ratio* represents market leverage, which is defined as total debt divided by the market value of assets. The variable *Log Debt* represents the log of total debt. The variable *Debt* ↑ is an indicator equal to one if net debt issuance is positive, that is, if total debt increased relative to the previous year. The variable *Capex* represents capital expenditures, and *Tot Inv* represents total investment, that is, the sum of capital expenditures, R&D, acquisitions, and advertising expenses. Panel B shows the results of the reduced-form estimation in which these outcomes are regressed directly on the *Predicted First Year* instrument. Control variables are those listed in Table II. Standard errors appear in parentheses and are clustered by firm. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Instrumental Variable Estimation					
	Leverage			Investment	
	(1)	(2)	(3)	(4)	(5)
	$\Delta \text{ Lev Ratio}$	$\Delta \text{ Log Debt}$	Debt ↑	Log Capx	Log Tot Inv
$\Delta \text{ Log } B\text{-}S \text{ Value}$	0.0964** (0.0484)	0.129** (0.0643)	0.350** (0.161)	−0.131 (0.184)	0.121 (0.215)
Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
F-stat (First Stage)	40.22	40.22	40.22	40.22	40.22
Observations	2,815	2,815	2,815	2,815	2,815
Panel B: Reduced-Form Estimation					
	Leverage			Investment	
	(1)	(2)	(3)	(4)	(5)
	$\Delta \text{ Lev Ratio}$	$\Delta \text{ Log Debt}$	Debt ↑	Log Capx	Log Tot Inv
Predicted First Year	0.0123** (0.00572)	0.0165** (0.00798)	0.0447** (0.0195)	−0.0167 (0.0235)	0.0155 (0.0278)
Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
R ²	0.11	0.13	0.09	0.82	0.78
Observations	2,815	2,815	2,815	2,815	2,815

Economic message. Even inside the fixed-value sample alone, where plan-type comparisons are irrelevant, the paper finds that more options raise risk and that leverage is the main channel.

5.3 Table VI: IV1 validity tests

Read:

- When predicted first years are split by whether option value actually increased, only the **increase** subgroup shows a strong volatility effect.
- Predicted first years are not associated with product-announcement changes, expansion announcements, debt maturing, or CEO turnover.

Economic message. The first instrument does not seem to be picking up unrelated firm cycles or CEO transitions.

Table VI
IV1: Validity Tests

Panel A reproduces the reduced-form specification from Panel B of Table IV, decomposing the predicted first year indicator into two variables. *Predicted First Year* ($\Delta BS\ Val > 0$) is equal to one if the observation corresponds to a predicted first year in which the value of the CEO's option pay has increased. *Predicted First Year* ($\Delta BS\ Val \leq 0$) is equal to one if the observation corresponds to a predicted first year in which the value of the CEO's option pay has weakly decreased. Panel B again reproduces the reduced-form specification from Panel B of Table IV, with different dependent variables. The variable *Product* represents the number of major product announcements the firm made during the year according to the CapitalIQ key developments database. The variable *Expansion* represents the number of business expansion announcements the firm made during the year. We consider CapitalIQ's coverage of a particular firm to begin with the firm's first recorded announcement of any type. The variable *% Debt Maturing* represents the percent of the firm's debt that was maturing in the year. We compute the percent of debt maturing in a year using the ratio of current liabilities to current liabilities plus long-term debt, as of the previous fiscal year-end. Financial firms (SIC 6000 to 6999) and regulated utilities (SIC 4800 to 4999) are excluded when this dependent variable is used. The variable *CEO turnover* is an indicator equal to one if it is the last year the CEO is at the firm (except if it was also the last year the firm was in the data). Control variables are those listed in Table II. Standard errors appear in parentheses and are clustered by firm. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Reduced-Form Estimation with Partitioned Instrument				
	Δ 12-Month Volatility		Δ 120-TD Volatility	
	(1)	(2)	(3)	(4)
Predicted First Year ($\Delta BS\ Val > 0$)	0.0198*** (0.00662)	0.0193*** (0.00672)	0.0204*** (0.00641)	0.0211*** (0.00640)
Predicted First Year ($\Delta BS\ Val \leq 0$)	0.00992 (0.00782)	0.00864 (0.00791)	0.00713 (0.00781)	0.00278 (0.00792)
Controls	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes
R^2	0.305	0.311	0.282	0.294
Observations	3,692	3,692	3,692	3,692

Panel B: Other Firm Cycles				
	(1) Δ Product	(2) Δ Expansion	(3) Δ % Debt Maturing	(4) CEO Turnover
Predicted First Year	-0.00944 (0.125)	-0.159 (0.158)	0.000670 (0.00514)	0.00578 (0.00874)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
R^2	0.025	0.022	0.031	0.028
Observations	3,478	3,478	2,795	2,950

Table VII
IV2: First Stage

This table shows the differential sensitivity of the option compensation of fixed-number and fixed-value CEOs to industry returns. Observations are at the CEO-year level. The sample is limited to CEOs who are on either fixed-number or fixed-value plans (excluding the first years of cycles). The variable *FN* is an indicator equal to one if the CEO is on a fixed-number plan. Industry returns are defined as the Fama-French (49) industry return of the CEO's firm in the 12 months preceding the option grant associated with the cycle. Other variables are defined as in Table II. The main effects of interaction terms are included in all specifications but not shown. Control variables are those listed in Table II. Standard errors appear in parentheses and are clustered by firm. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Δ Log B-S Value		Δ Log Delta		Δ Log Vega	
	(1)	(2)	(3)	(4)	(5)	(6)
FN $\times$ Ind Return	0.559*** (0.0473)	0.548*** (0.0474)	0.630*** (0.0878)	0.599*** (0.0892)	0.546*** (0.114)	0.510*** (0.108)
Controls	No	Yes	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.182	0.261	0.082	0.281	0.153	0.286
Observations	3,535	3,535	3,535	3,535	3,535	3,535

5.4 Table VII to Table IX: IV2 cross-validates the main result with a totally different source of variation

Read:

- **Table VII:** the interaction of fixed-number status with industry returns strongly predicts changes in new option value; the main interaction coefficient is about **0.55**.
- **Table VIII:** the IV coefficients on volatility are about **0.086** for 12-month volatility and **0.144** for 120-trading-day volatility in the controlled specifications. These map into a **2.8% to 4.2%** increase in volatility from a 10% increase in new options.
- **Table IX:** leverage again rises sharply. A 10% increase in options raises the market leverage ratio by about **0.007**, book debt by about **1.3%**, and the probability of positive net issuance by about **3.1 percentage points**.

Economic message. The second IV strategy lands in almost the same place as the first one. That convergence is one of the strongest features of the paper.

5.5 Table X to Table XIII: robustness and heterogeneity

Read:

- **Table X:** allowing highly nonlinear responses to industry returns and letting those nonlinearities vary with observables leaves the IV2 estimates broadly intact.

Table VIII
IV2: Volatility

Panel A shows IV estimation results, where $\Delta \text{Log B-S Value}$ is instrumented using $FN \times \text{Ind Return}$, as defined in Table VII. Observations are at the CEO-year level. The sample is limited to CEOs who are on either fixed-number or fixed-value plans (excluding the first years of cycles). All other variables are as defined in Table IV. Panel B shows the results of the reduced-form estimation in which these outcomes are regressed directly on the $FN \times \text{Ind Return}$ instrument. The main effects of interaction terms are included in all specifications but not shown. Standard errors appear in parentheses and are clustered by firm. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Instrumental Variable Estimation				
	Δ 12-Month Volatility		Δ 120-TD Volatility	
	(1)	(2)	(3)	(4)
$\Delta \text{ Log B-S Value}$	0.0832** (0.0424)	0.0864** (0.0432)	0.138*** (0.0457)	0.144*** (0.0466)
Controls	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes
<i>F</i> -stat (First Stage)	140.04	133.82	140.04	133.82
Observations	3,535	3,535	3,535	3,535
Panel B: Reduced-Form Estimation				
	Δ 12-Month Volatility		Δ 120-TD Volatility	
	(1)	(2)	(3)	(4)
$FN \times \text{Ind Return}$	0.0465** (0.0200)	0.0474** (0.0201)	0.0773*** (0.0251)	0.0788*** (0.0251)
Controls	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes
<i>R</i> ²	0.260	0.263	0.257	0.261
Observations	3,535	3,535	3,535	3,535

Table X
Interacted Controls and Nonlinearity

This table reestimates Table VIII with additional controls. First, rather than controlling linearly for the main effect of industry returns, we include a five-degree polynomial of industry returns. Second, in Columns (3) and (6) we allow the five-degree polynomial to interact with all of the control variables listed in Table II to control for the fact that firms with different observable characteristics might respond differently (and nonlinearly) to industry returns. Across all specifications with a five-degree polynomial of industry returns, the excluded instrument is the interaction between the five-degree polynomial of industry returns and the fixed-number plan indicator variable. Standard errors appear in parentheses and are clustered by firm. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Δ 12-Month Volatility			Δ 120-TD Volatility		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \text{ Log B-S Value}$	0.0866** (0.0415)	0.0882** (0.0425)	0.107*** (0.0414)	0.127*** (0.0427)	0.128*** (0.0439)	0.0917** (0.0447)
Controls	No	Yes	Yes	No	Yes	Yes
Poly. Ind. Returns	Yes	Yes	Yes	Yes	Yes	Yes
Controls $\times$ Poly. Ind. Returns	No	No	Yes	No	No	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>F</i> -stat (First Stage)	41.58	37.77	37.60	41.58	37.77	37.60
Observations	3,535	3,535	3,535	3,535	3,535	3,535

Table IX
IV2: Other Outcomes

Panel A shows instrumental variable estimation results, where $\Delta \text{Log B-S Value}$ is instrumented using $FN \times \text{Ind Return}$, as defined in Table VII. Observations are at the CEO-year level. The sample is limited to CEOs who are on either fixed-number or fixed-value plans (excluding the first years of cycles). Financial firms (SIC 6000 to 6999) and regulated utilities (SIC 4800 to 4999) are excluded. All other variables are as defined in Table V. Panel B shows the results of the reduced-form estimation in which these outcomes are regressed directly on the $FN \times \text{Ind Return}$ instrument. The main effects of interaction terms are included in all specifications but not shown. Control variables are those listed in Table II. Standard errors appear in parentheses and are clustered by firm. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Instrumental Variable Estimation					
	Leverage			Investment	
	(1) Δ Lev Ratio	(2) Δ Log Debt	(3) Debt $\uparrow$	(4) Log Capx	(5) Log Tot Inv
$\Delta \text{ Log B-S Value}$	0.0717** (0.0315)	0.129** (0.0626)	0.310** (0.135)	0.307 (0.191)	0.196 (0.235)
Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
<i>F</i> -stat (First Stage)	122.83	122.83	122.83	122.83	122.83
Observations	2,930	2,930	2,930	2,930	2,930
Panel B: Reduced-Form Estimation					
	Leverage			Investment	
	(1) Δ Lev Ratio	(2) Δ Log Debt	(3) Debt $\uparrow$	(4) Log Capx	(5) Log Tot Inv
$FN \times \text{Ind Return}$	0.0427** (0.0179)	0.0718** (0.0342)	0.173** (0.0741)	0.171 (0.106)	0.109 (0.132)
Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
<i>R</i> ²	0.20	0.13	0.08	0.84	0.75
Observations	2,930	2,930	2,930	2,930	2,930

Table XI
IV2: Placebo Test

Panel A of this table shows reduced-form regression results where the outcomes in Table IX are regressed directly on the $FN \text{ Placebo} \times \text{Ind. Return}$ placebo instrument. The placebo sample is restricted to CEOs receiving option pay who are not currently on a cycle, but were on a fixed-number or fixed-value cycle in some other year (in the past or future). $FN \text{ Placebo}$ is an indicator variable equal to one if the CEO was on a fixed-number cycle in some other year. The main effects of the interaction terms are included in all specifications but not shown. In Panel B of this table, the placebo sample is instead limited to firms that discontinued granting options altogether (for all top executives) in the fiscal year before, the fiscal year of, or the fiscal year after FAS123r went into effect for the firm (FAS123r applied to the first fiscal year starting after June 15, 2005). The sample only includes the post-FAS123r period. In this case, $FN \text{ Placebo}$ is an indicator variable equal to one if the CEO was on a fixed-number cycle prior to the firm ceasing the use of option pay and zero if the CEO was on a fixed-value cycle. Control variables are those listed in Table II. Standard errors appear in parentheses and are clustered by firm. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Placebo Test							
	Volatility		Leverage		Investment		
	(1) Δ 12-Month Vol	(2) Δ 120 TD Vol	(3) Δ Lev Ratio	(4) Δ Log Debt	(5) Debt $\uparrow$	(6) Log Capx	(7) Log Tot Inv
$FN \text{ Placebo} \times \text{Ind Return}$	−0.000489 (0.0175)	−0.0164 (0.0164)	−0.00615 (0.0149)	0.0118 (0.0240)	0.0198 (0.0499)	−0.0493 (0.0708)	0.0284 (0.0913)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>R</i> ²	0.29	0.29	0.12	0.13	0.08	0.85	0.75
Observations	6,098	6,098	5,647	5,647	5,647	5,647	5,647
Panel B: Placebo Test with Firms that Discontinued Options Surrounding FAS123r							
	Volatility		Leverage		Investment		
	(1) Δ 12-Month Vol	(2) Δ 120-TD Vol	(3) Δ Lev Ratio	(4) Δ Log Debt	(5) Debt $\uparrow$	(6) Log Capx	(7) Log Tot Inv
$FN \text{ Placebo} \times \text{Ind Return}$	−0.0271 (0.0493)	−0.0192 (0.0461)	0.0196 (0.0389)	−0.0652 (0.0560)	−0.167 (0.175)	−0.0464 (0.237)	−0.210 (0.209)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>R</i> ²	0.55	0.67	0.18	0.15	0.14	0.82	0.74
Observations	657	657	551	551	551	551	551

- **Table XI:** placebo versions of the instrument among firms that stop granting options do not generate similar effects on volatility, leverage, or investment.
- **Table XII:** the volatility response is much larger when new options are a high share of all unexercised options. For IV1, the reduced-form coefficient rises from about **0.012** in the low group to **0.031** in the high group; for IV2 it rises from about **0.029** to **0.156**.
- **Table XIII:** the results survive controls for total compensation and other compensation components, no-restricted-stock subsamples, two-year-cycle subsamples, one-grant-per-year subsamples, and predicted-grant-date tests.

Economic message. The core finding is not fragile. It survives an extensive battery of specification, placebo, and subsample checks.

Table XII
Heterogeneity

This table reestimates the baseline reduced-form specifications from Column (3) of Tables IV and VIII, Panel B. The change in 120-trading day volatility is regressed on *Pred First Year* in the leftmost columns and on *FN × Ind Return* in the rightmost columns. In Panel A, the sample is split into terciles based upon the ratio of the value of new options to all unexercised options as of the grant date of the new options. The High columns are the top tercile sample, and the Low columns are the bottom tercile sample. The *p*-value row represents the *p*-value of the test that the coefficients in the High and Low column are equal. In Panel B, the sample is split based upon whether the company is in the finance (SIC 6000 to 6999) or high-tech sectors ("high-tech" in Fama-French five-industry classification). Control variables are those listed in Table II. Standard errors appear in parentheses and are clustered by firm. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Percent New Options				
DEP VAR: Δ 120 TD VOL	Instrument = Pred First Year		Instrument = FN × Ind Return	
	(1) Low	(2) High	(3) Low	(4) High
Instrument	0.0124* (0.00738)	0.0313*** (0.00906)	0.0291 (0.0456)	0.156*** (0.0402)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
<i>p</i> -value	0.044	0.044	0.004	0.004
<i>R</i> ²	0.359	0.321	0.313	0.283
Observations	1,230	1,229	1,178	1,177

Panel B: Finance and High Tech				
DEP VAR: Δ 120 TD VOL	Instrument = Pred First Year		Instrument = FN × Ind Return	
	(1) No Fin/Tech	(2) Fin/Tech	(3) No Fin/Tech	(4) Fin/Tech
Instrument	0.0108* (0.00617)	0.0201** (0.00869)	0.0536* (0.0319)	0.0955** (0.0413)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
<i>p</i> -value	0.336	0.336	0.272	0.272
<i>R</i> ²	0.269	0.338	0.254	0.299
Observations	2,524	1,168	2,364	1,171

Table XIII
Robustness

This table reestimates the baseline instrumental variable specifications from Column (3) of Tables IV and VIII, Panel A. In both panels, Δ *Log B-S Value* is instrumented using the *Predicted First Year* indicator in odd-numbered columns and *FN × Ind Return* in even-numbered columns. In the first two columns of Panel A, controls for the annual change in log total compensation (the sum of the grant date values of salary, bonus, restricted stock grants, option grants, and all other compensation) are added to the baseline instrumental variable specifications. In Columns (3) and (4), separate controls for the annual changes in the grant date values of the logarithms of salary, bonus, restricted stock grants, and all other compensation are added to the baseline instrumental variable specifications. In Columns (5) and (6), the sample is restricted to CEO-years in which no restricted stock is granted. In the first two columns of Panel B, the sample is restricted to two-year cycles. In Columns (3) and (4), the sample is restricted to the set of CEO-years in which only one option grant is awarded. In Columns (5) and (6), rather than computing volatility following the reported grant date, we compute volatility following a predicted grant date (using the same month and day of the previous year's reported grant date). The main effects of interaction terms are included in all specifications but not shown. Control variables are those listed in Table II. Standard errors appear in parentheses and are clustered by firm. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Other Compensation						
DEP VAR: Δ 120 TD VOL	Total Comp Control		Component Controls		No Restricted Stock	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Log B-S Value	0.171*** (0.0618)	0.189*** (0.0587)	0.0937*** (0.0354)	0.143*** (0.0463)	0.0715** (0.0348)	0.149*** (0.0513)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Instrument	IV1	IV2	IV1	IV2	IV1	IV2
<i>F</i> -stat (First Stage)	33.35	98.00	65.08	135.98	49.03	103.85
Observations	3,692	3,535	3,692	3,535	2,212	2,425

Panel B: Additional Tests						
DEP VAR: Δ 120 TD VOL	2-Yr Cycles		1 Grant/Yr		Predicted Grant Date	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Log B-S Value	0.0991*** (0.0356)	0.150*** (0.0552)	0.115*** (0.0404)	0.110** (0.0488)	0.104** (0.0461)	0.165*** (0.0549)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Instrument	IV1	IV2	IV1	IV2	IV1	IV2
<i>F</i> -stat (First Stage)	58.61	69.69	52.70	126.27	52.51	156.75
Observations	3,326	2,147	3,271	3,008	3,468	3,228

6 Short summary

- Shue and Townsend study the causal effect of CEO option grants on risk-taking using the institutional structure of multiyear compensation plans.
- They use two IV designs: predicted first years of fixed-value cycles and the interaction of fixed-number plans with industry returns.
- Both designs imply that a 10% increase in new option grants raises firm equity volatility by about 2.8% to 4.2%.
- The main channel is higher leverage, while investment responses are small.

- The results are strongest when the new option grant is large relative to the CEO's existing option portfolio.

7 Conclusion

7.1 Core conclusion

The paper provides unusually clean causal evidence that increases in CEO option grants raise corporate risk-taking. The key contribution is not just the sign of the effect, but the design: by exploiting quasi-random timing generated by multiyear pay plans, the paper gets around the usual objection that boards raise option pay exactly when they want more risk.

7.2 Economic intuition

At the grant sizes typically observed in the data, the convexity effect of new at-the-money options dominates any offsetting magnification effect from higher delta. CEOs respond by increasing firm risk, and the fastest way to do that in the data is to increase leverage.

7.3 Takeaway

For empirical corporate finance, the paper is important because it turns a classic agency-theory prediction into a convincing causal estimate. For boards and policy makers, the message is practical: moderate increases in option pay do not simply motivate effort; they also change the firm's risk profile, mainly through capital-structure choices.